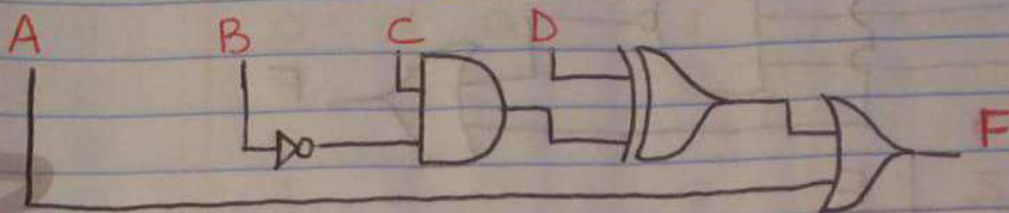


Lista de Exercícios 4

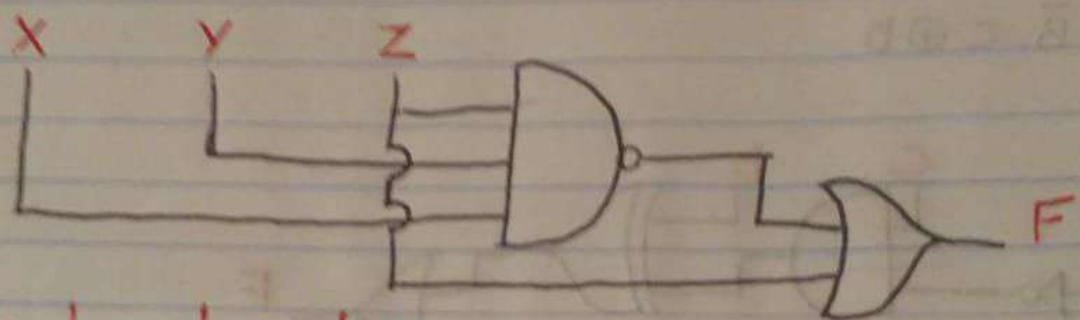
1-a) $F = A + \bar{B} \cdot C \oplus D$



A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

$$\bar{C} \cdot \bar{B} + D + C \cdot \bar{A}$$

b) $\overline{X} \cdot Y \cdot Z + Z = F$



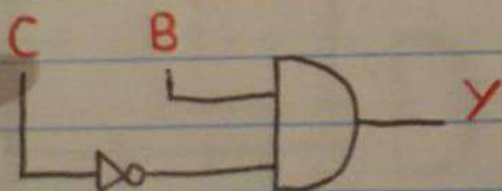
X	Y	Z	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

2- Determine o circuito para as tabelas verdade abaixo:

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	0	0
11	1	1	0	0
10	0	0	0	0

$$Y = \bar{C} \cdot B$$

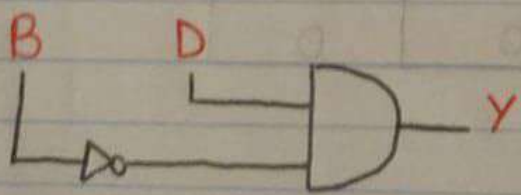


b)

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	0	0	0	0
10	0	1	1	0

$$Y = D \cdot \bar{B}$$

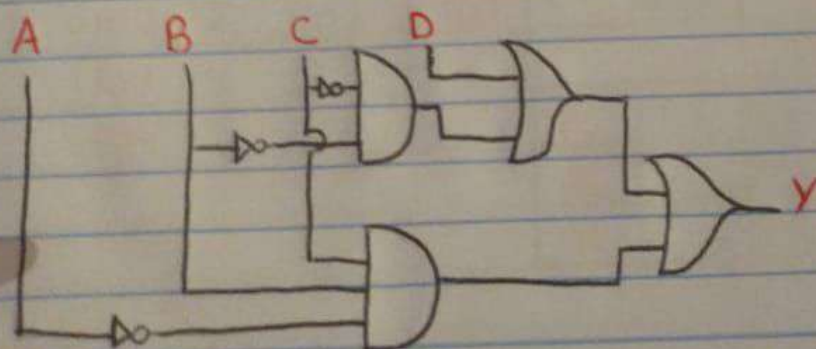


c)

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

AB \ CD	00	01	11	10
00	1	1	1	0
01	0	1	1	1
11	0	1	1	0
10	1	1	1	0

$$Y = \bar{C} \cdot \bar{B} + D + C \cdot \bar{A} \cdot B$$



3 - Derivado: Abaixo encontra-se o flip-flop JK, monta a tabela verdade para as entradas J e K com a saída Q.

J	K	CLK	Q	$\overline{Q}$
0	0	↑	Anterior	$\overline{\text{Anterior}}$
0	1	↑	0	1
1	0	↑	1	0
1	1	↑	$\overline{\text{Anterior}}$	Anterior
X	X	0	Anterior	$\overline{\text{Anterior}}$
X	X	1	Anterior	$\overline{\text{Anterior}}$
X	X	↑	Anterior	$\overline{\text{Anterior}}$